

Predicting Pitting Potential

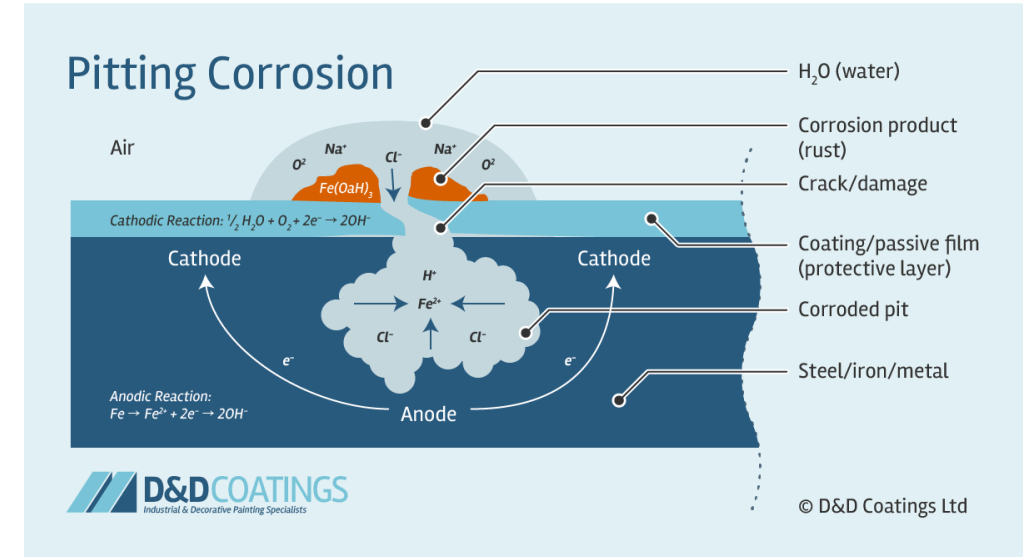
Corrosion-Informed Learning with TabICL

Bringing corrosion knowledge into tabular machine learning, from experimental data to model evaluation.

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Pitting Potential (EPIT)

A Measure of Pitting Resistance



- **Pitting corrosion:** local breakdown of the protective passive film allows pits to form.
- **EPIT:** the electrochemical potential at which breakdown or sustained pitting is detected during a test.
- **Higher EPIT** → **greater resistance to pit initiation**, under comparable conditions.
- EPIT depends on the **alloy, material state, environment, and test procedure**.

From Alloy Composition and Test Conditions to EPIT

760 rows · 21 input variables · 1 prediction target

- **Composition:** 17 element weight percentages, including Fe, Cr, Ni, Mo, and W.
- **Test conditions:** temperature, chloride concentration, pH, and experimental method.
- **Target:** average pitting potential, measured in **mV vs SCE**.

	A	B	C	D	E	F	G	H	I
1	Composition, wt.% Fe	Composition, wt.% Cr	Composition, wt.% Ni	...	Test Temp. oC	[Cl-] M	[Cl-] pH	[Cl-] Test Method	Epit, mV (SCE) Avg.
2									
3	69.705	18	10		50	0.005133497	7.8	Potentiodynamic Polarization wit	287
4	69.705	18	10		50	0.008555828	7.8	Potentiodynamic Polarization wit	260
5	69.705	18	10		50	0.017111655	7.8	Potentiodynamic Polarization wit	218
6	69.705	18	10		50	0.085558276	7.8	Potentiodynamic Polarization wit	140
7	69.705	18	10		50	0.546	8.2	Potentiodynamic Polarization wit	94
8	67.535	16	11		50	0.005133497	7.8	Potentiodynamic Polarization wit	344
9	...								
10	69.819	18.92	8.51		25	0.5	-0.301	Potentiodynamic Polarization	265.5
11	69.595	18.87	8.59		25	0.5	-0.301	Potentiodynamic Polarization	328.5
12	69.601	18.8	8.63		25	0.5	-0.301	Potentiodynamic Polarization	532.3
13	58.389	17.38	20.15		25	0.6	7.68	Following pretreatment for 5 min	-65
14	57.048	17.68	20.43		25	0.6	7.68	Following pretreatment for 5 min	-25
15	...								
16	68.534	16.3	14.4		60	0.1	7	Polarization curves were obtaine	77.7
17	66.475	16.3	14		60	0.1	7	Polarization curves were obtaine	214.3
18	63.874	16.5	14.1		60	0.1	7	Polarization curves were obtaine	297.1
19	66.475	16.3	14		60	0.1	1	Polarization curves were obtaine	58.8377724
20	63.874	16.5	14.1		60	0.1	1	Polarization curves were obtaine	121.3075061
21	60.344	16.17	14.1		60	0.1	1	Polarization curves were obtaine	208.4745763
22	56.644	16.22	14		60	0.1	1	Polarization curves were obtaine	252.0581114
23	66.475	16.3	14		60	0.1	1	Polarization curves were obtaine	122.0007624

How a Conventional Machine-Learning Model Works

- A model is trained directly on the available dataset.
- During training, it compares predictions with known target values and adjusts its parameters.
- The result is one model fitted specifically to that dataset.
- A different target or dataset normally requires new training.

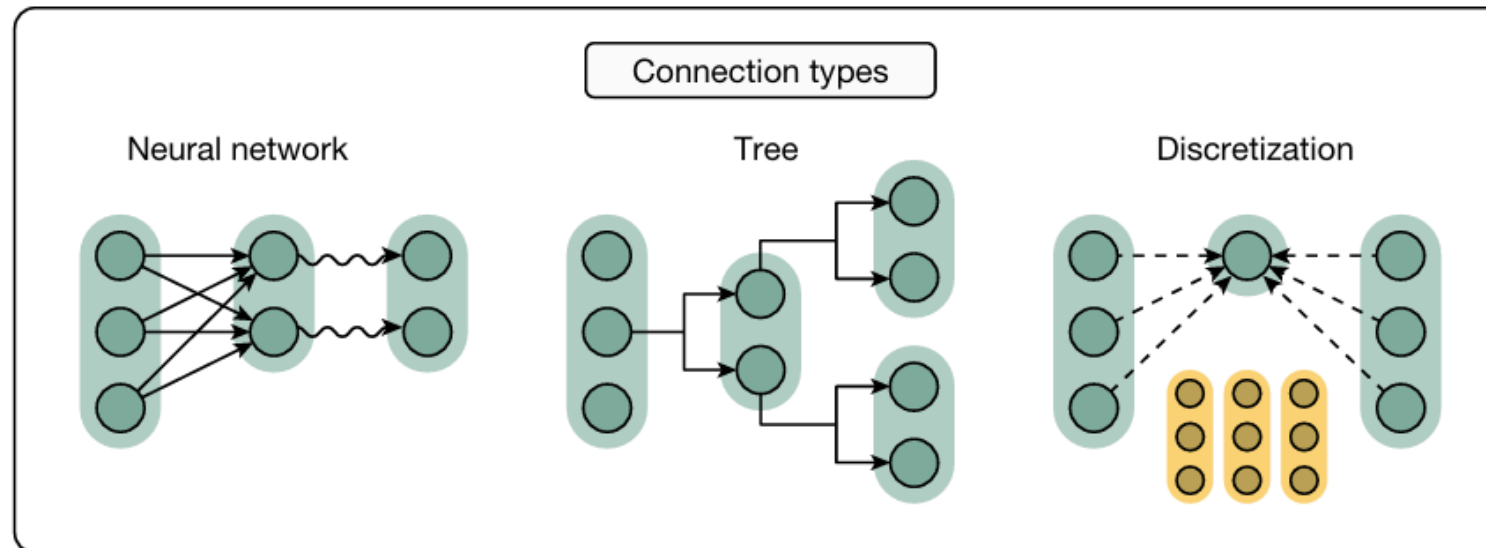
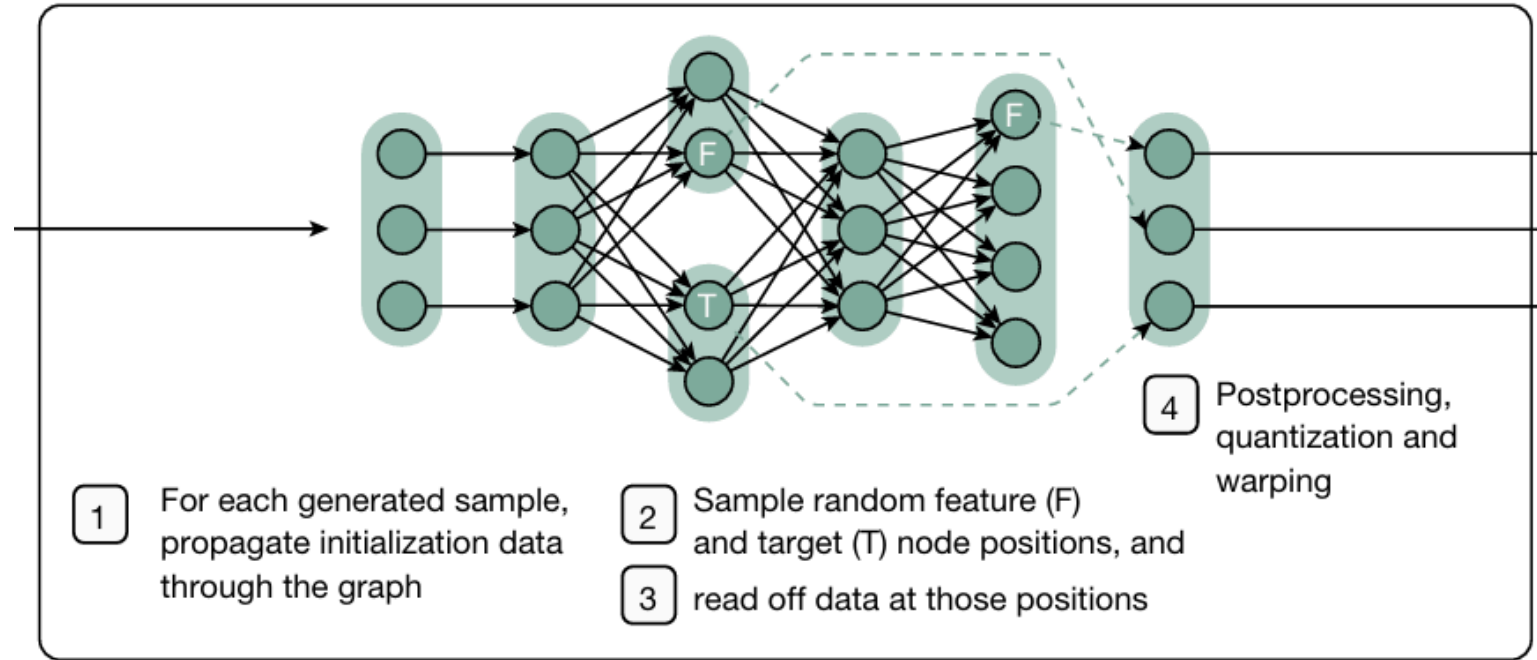
$$(x, y) \xrightarrow{\text{fitting}} \text{fitted model}, \quad (\text{fitted model}, x^*) \longrightarrow y^*$$

A Model Pretrained to Learn from Tables

- **Tabular foundation models such as TabPFN and TabICL are gaining attention, often outperforming conventional models on benchmark datasets.**
- TabICL is pretrained before seeing the real EPIT task.
- Pretraining uses many automatically generated synthetic tables.
- These tables have different relationships between features and target.
- **The model still needs labeled EPIT examples.** It uses them as context during prediction rather than to retrain its parameters.

$$\underbrace{D_1, D_2, \dots, D_N}_{\text{Synthetic tables}} \xrightarrow{\text{pretraining}} \underbrace{\text{TabICL}_\phi}_{\text{Learned weights } \phi}$$
$$(x, y, x^*) \xrightarrow{\text{pretrained TabICL}} q_\phi(y^* \mid x^*, x, y)$$

b Build computational graph and graph structure



How TabCL Builds a Prediction

TabCL uses **attention** to learn how to combine information from features and examples.

- **Column:** use other values in the column to give each value context.
- **Row:** combine its features into a representation of the whole row.
- **Examples:** use rows with known targets to guide predictions for new rows.
- **Training:** compare predictions with the generated targets and update the model's parameters.

The Case for an Informed Prior

- Generic TablCL sees anonymous columns such as $X_1, X_2, \dots$
- It does not know which feature represents chromium, chloride, temperature, or test method.
- Its synthetic target has no pitting-potential interpretation.
- It learns general statistical patterns, but its synthetic training contains no deliberate corrosion structure.

Central project idea: Make some synthetic training tasks more relevant to EPIT by including alloy compositions, test conditions, and corrosion-inspired relationships.

Our Corrosion-Informed Implementation

Corrosion-relevant input distributions

- **Alloy compositions:** sample recorded Fe and Ni-Cr-Mo compositions and introduce random variations.
- **Test conditions:** sample recorded combinations of temperature, chloride concentration, pH, and test method.

Corrosion-informed target formulas

- A random SCM has no built-in reason to link higher chloride with lower pitting resistance.
- We add formulas that calculate a target score from **alloy composition and test conditions**.
- These formulas express corrosion tendencies from the literature: **material protection raises the score; environmental aggressiveness lowers it.**

Seven rules

- Linear PREN-inspired
- Cr-Mo/W synergy
- Threshold and saturation
- Separate environmental effects
- Coupled breakdown
- Fe/Ni chromium threshold
- Method-aware

Example: Separate environmental effects

$$y_{\text{rule}} = aP - bC - cH - dI - eA$$

- P : material protection, based on $\text{Cr} + 3.25(\text{Mo} + 0.55 \text{ W})$.
 - C : chloride aggressiveness, based on log concentration.
 - H : a penalty that increases with temperature.
 - I : an additional penalty when temperature and chloride are both high.
 - A : a penalty that increases as pH decreases.
1. Fit formula weights to match measured EPIT rankings in development data.
 2. Choose one of seven formulas per synthetic table, favoring those that perform better on validation data.
 3. Standardize and combine the chosen rule's target value with the original generator's target value.

The coefficients a, b, c, d, e control the strength of each contribution.

Mean Validation Spearman Scores of Each Rule

Rule	Score
Linear PREN-inspired	0.5174
Cr-Mo/W synergy	0.5583
Threshold and saturation	0.5181
Separate environmental effects	0.5850
Coupled breakdown	0.5525
Fe/Ni chromium threshold	0.5252
Method-aware	0.5999

- Applied each rule to real corrosion rows excluded from fitting.
- Calculated the Spearman correlation between rule scores and measured EPIT values.
- Averaged the correlations across five folds.

Optuna: Best Trial and Parameters

- Optuna compared settings using **1,000-step training runs**.
- **Best completed trial: 56.**
- **Mean validation Spearman: 0.6903**, averaged across five development folds.

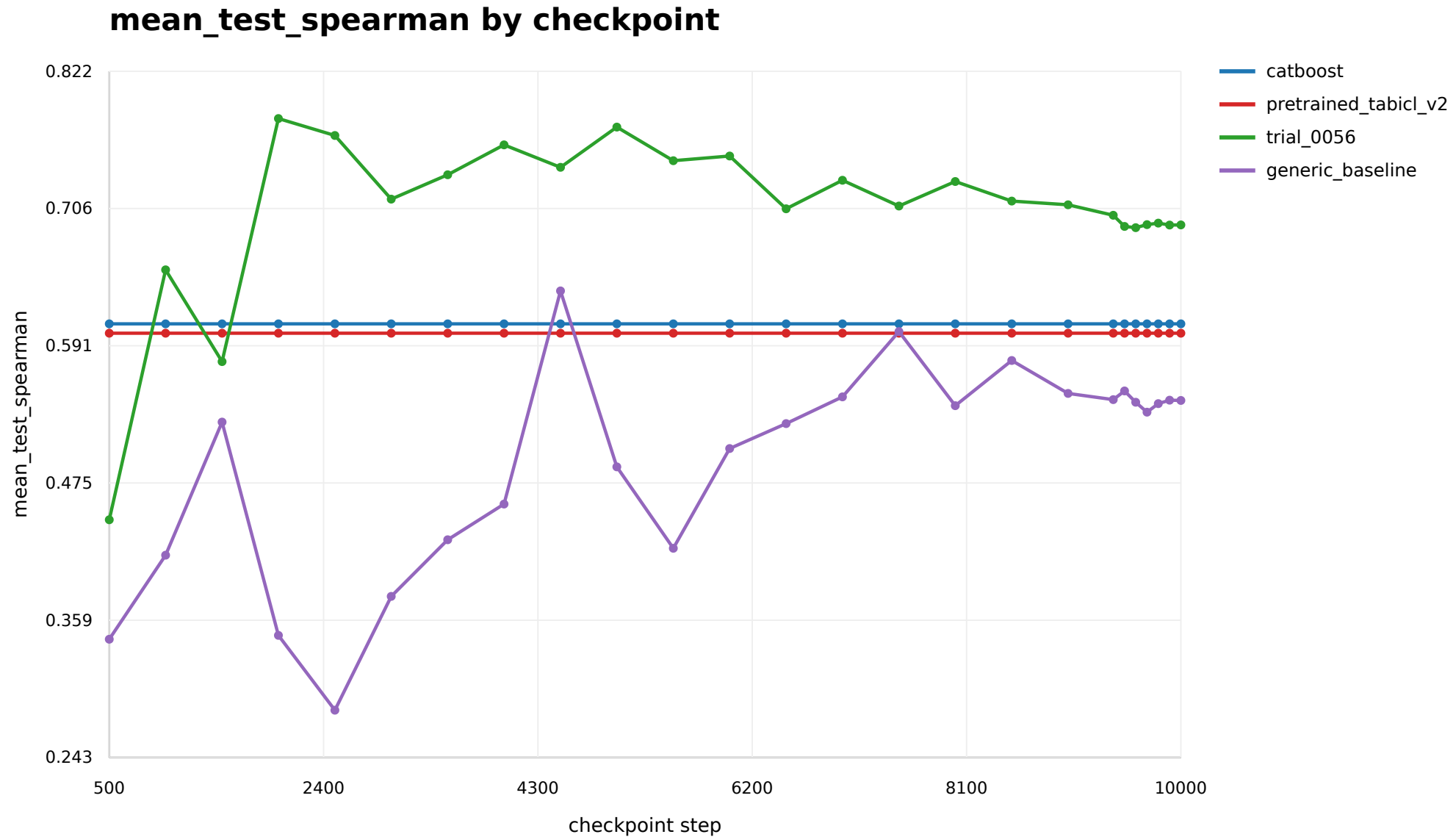
Parameter	Selected value
Share of corrosion-informed tables	75% informed, 25% original generator
Composition variation strength	0.1028
Generator within informed tables	50% MLP, 50% tree
Rule weight in the combined target	20.48% rule, 79.52% original generator

Comparison Models and Testing

Model	Training
Generic TabICL	Trained from scratch: V2 architecture, V1 data generator.
Pretrained TabICL V2	Authors' pretrained weights: V2 architecture and V2 data generator.
CatBoost	Trained directly on real EPIT data.

- Split development folds and the final-test set mainly by **alloy composition** to reduce data leakage.
- Used development rows as **labeled context** and final-test rows as **queries with unseen EPIT targets**.
- Final-test targets were **not used to fit formulas, update model parameters, or select settings**.

Model Comparison: Spearman Correlation



Recap and Next Steps

- The Approach Shows Promise
- Test Broader Applicability
- Identify What Drives the Improvement
- Explore TablCL V2 Generation
- Add Magpie descriptors

Appendix

Appendix · 1. Linear PREN-Inspired

$$y = aP - bE - cI_{\text{MC}}$$

- **P — material protection:** based on $\text{Cr} + \alpha(\text{Mo} + \eta \text{W})$.
- **E — environmental aggressiveness:** combines chloride, temperature, and pH.
- **I_{MC} — material-chloride interaction:** increases when protection is weak and chloride is high.

Idea: Cr, Mo, and W improve pitting resistance; aggressive conditions and weak protection lower the target score.

Appendix · 2. Cr-Mo/W Synergy

$$y = aP_{\text{syn}} - bE - cI_{\text{MC}} - dI_{\text{TC}}$$

- P_{syn} — **cooperative protection**: based on $\text{Cr} + \alpha q + \beta \sqrt{\text{Cr} q}$, with $q = \text{Mo} + \eta \text{W}$.
- E **and** I_{MC} : environmental aggressiveness and the weak-protection/chloride interaction.
- I_{TC} : an extra penalty when temperature and chloride are both high.

Idea: Chromium and Mo/W can reinforce each other's protective effect; their joint benefit exceeds the additive contribution.

Appendix · 3. Threshold and Saturation

$$y = aP_{\text{sat}} - bE - cI_{\text{MC}} - dI_{\text{TC}}$$

- P_{sat} — **nonlinear protection**: based on $\sigma((\text{Cr} - \theta)/s) + \ln(1 + q/q_0)$, with $q = \text{Mo} + \eta W$.
- σ is a smooth sigmoid: protection rises around chromium threshold θ ; the logarithm gives diminishing Mo/W returns.
- $E, I_{\text{MC}}, I_{\text{TC}}$: environment, weak-protection/chloride, and temperature/chloride penalties.

Idea: Chromium protection rises around a passivation threshold, while additional Mo/W gives diminishing benefits.

Appendix · 4. Separate Environmental Effects

$$y = aP - bC - cH - dI_{TC} - eA$$

- P : material protection based on Cr, Mo, and W.
- C **and** H : separate chloride and high-temperature penalties.
- I_{TC} : additional penalty when chloride and temperature are both high.
- A : acidic-pH penalty.

Idea: Give each environmental effect its own weight so chloride, temperature, their interaction, and acidity can contribute differently.

Appendix · 5. Coupled Breakdown

$$y = aP - bB - cA, \quad B = \sigma(G - \gamma M)$$

- **P and M :** protection and its underlying standardized Cr-Mo/W material signal.
- **G :** combined aggressiveness from chloride, high temperature, and their interaction.
- **B :** smooth breakdown response; it grows as aggressiveness exceeds material protection.
- **A :** acidic-pH penalty.

Idea: The same environment causes more breakdown in a weakly protected alloy. Stronger material protection shifts the breakdown threshold.

Appendix · 6. Fe/Ni Chromium Threshold

$$y = aP_f - bC - cH - dI_{TC} - eA$$

- P_f — **family-specific protection**: based on $\sigma((Cr - \theta_f)/s) + \ln(1 + q/q_0)$, with $q = Mo + \eta W$.
- θ_f : a lower chromium threshold for Fe alloys and a higher one for Ni-Cr-Mo alloys.
- C, H, I_{TC}, A : separate chloride, high-temperature, joint, and acidic-pH penalties.

Idea: Fe and Ni-Cr-Mo alloys need different chromium thresholds to develop protection.

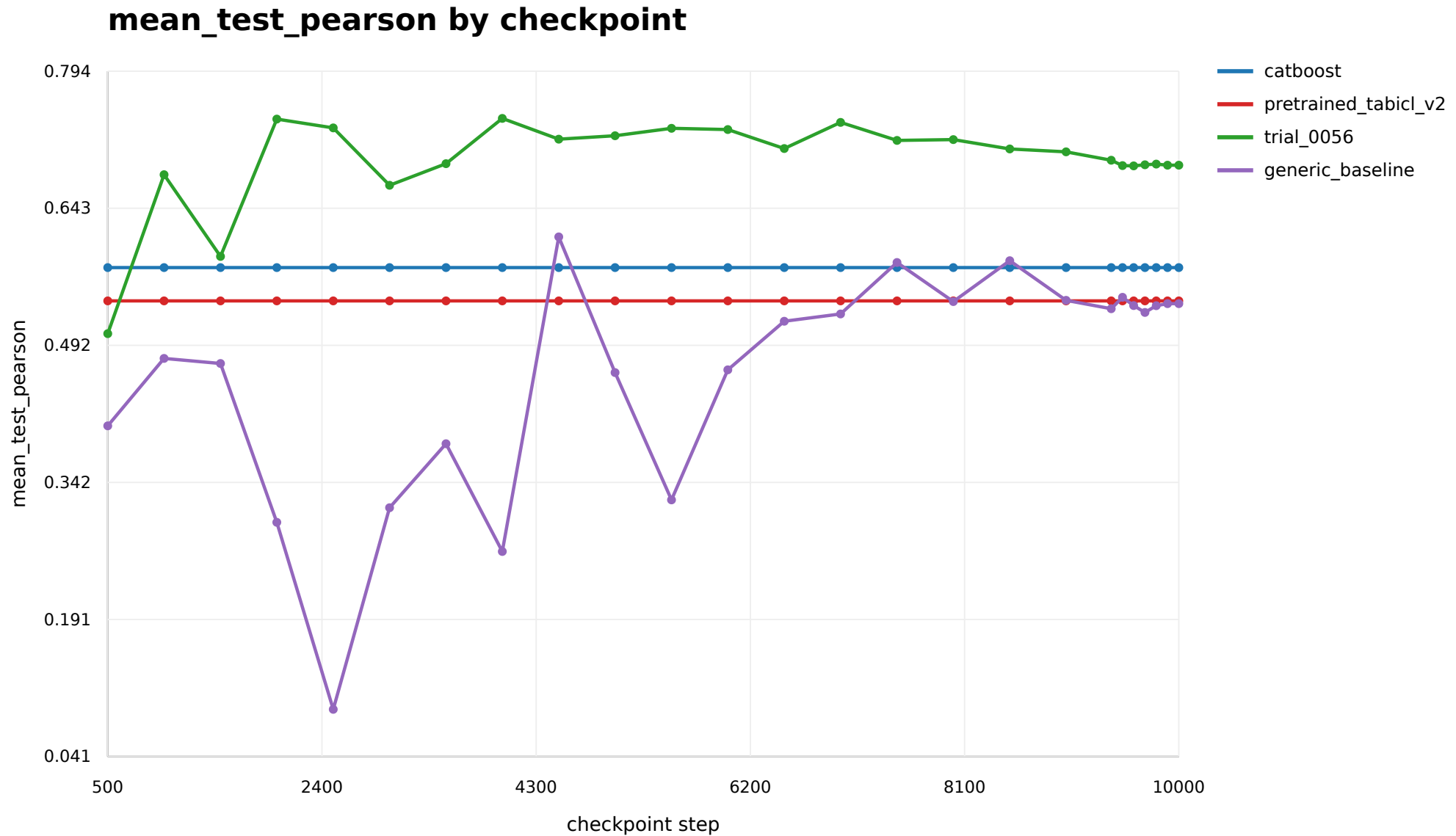
Appendix · 7. Method-Aware

$$y = aP_{\text{syn}} - bE - cI_{\text{MC}} - dI_{\text{TC}} + eD_m$$

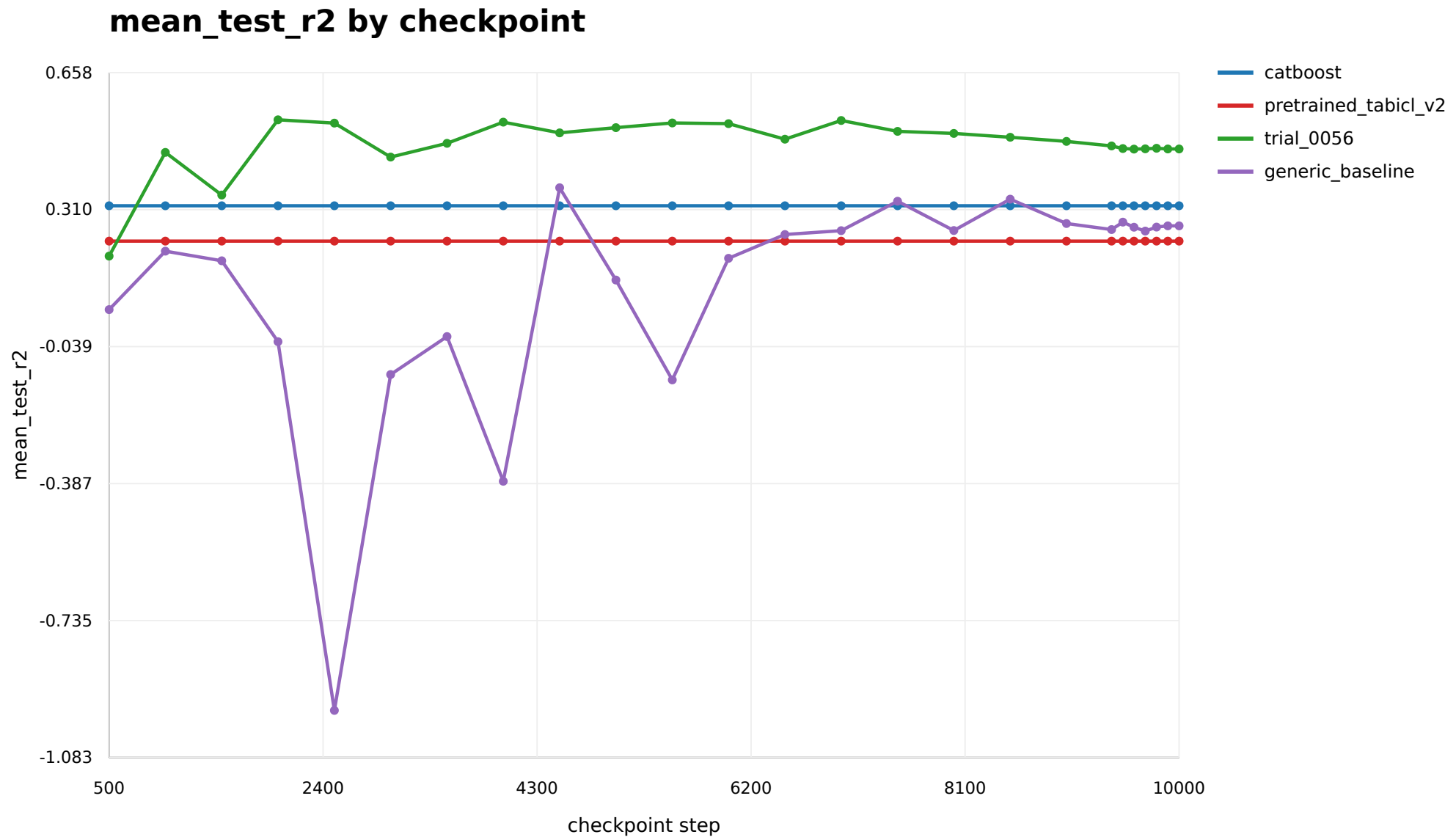
- P_{syn} : Cr-Mo/W synergy protection.
- $E, I_{\text{MC}}, I_{\text{TC}}$: environmental and interaction penalties.
- D_m : an offset associated with the test method; rows measured by the same method share it.
- Synthetic tables receive newly sampled method offsets.

Idea: Experimental methods can shift the target score even when alloy composition and environmental conditions are similar.

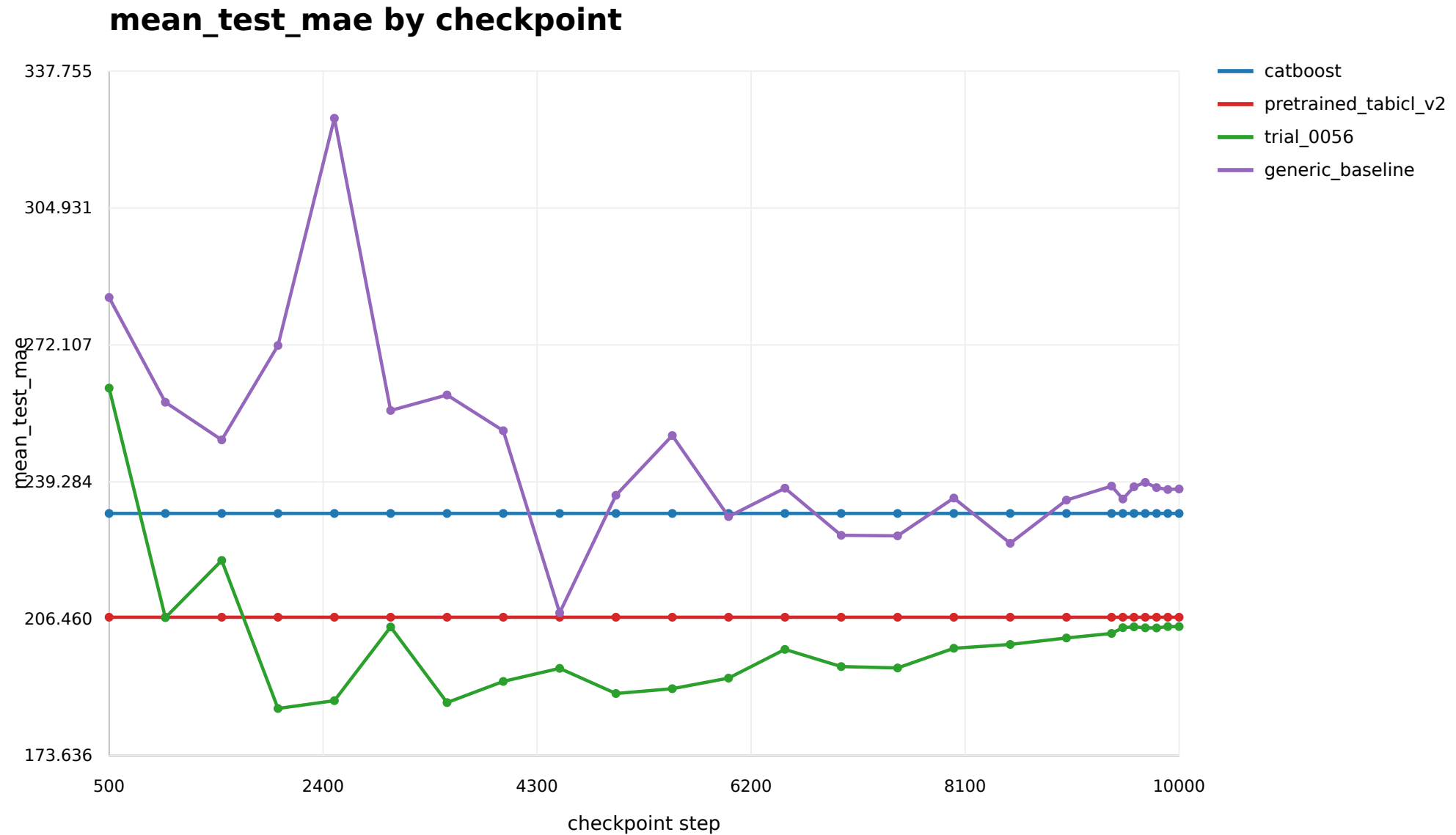
Appendix - 8. Pearson Correlation



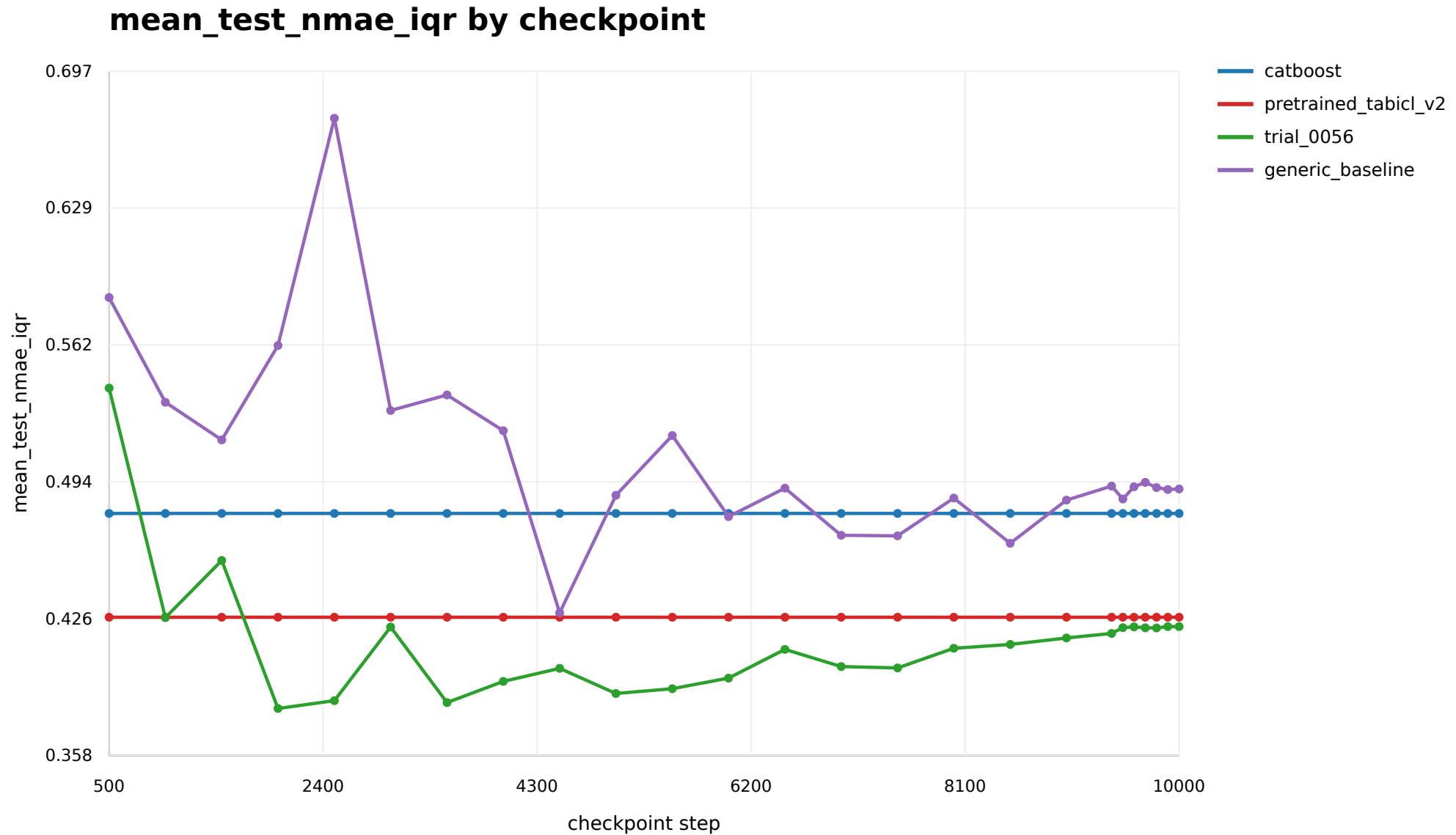
Appendix - 9. R² Score



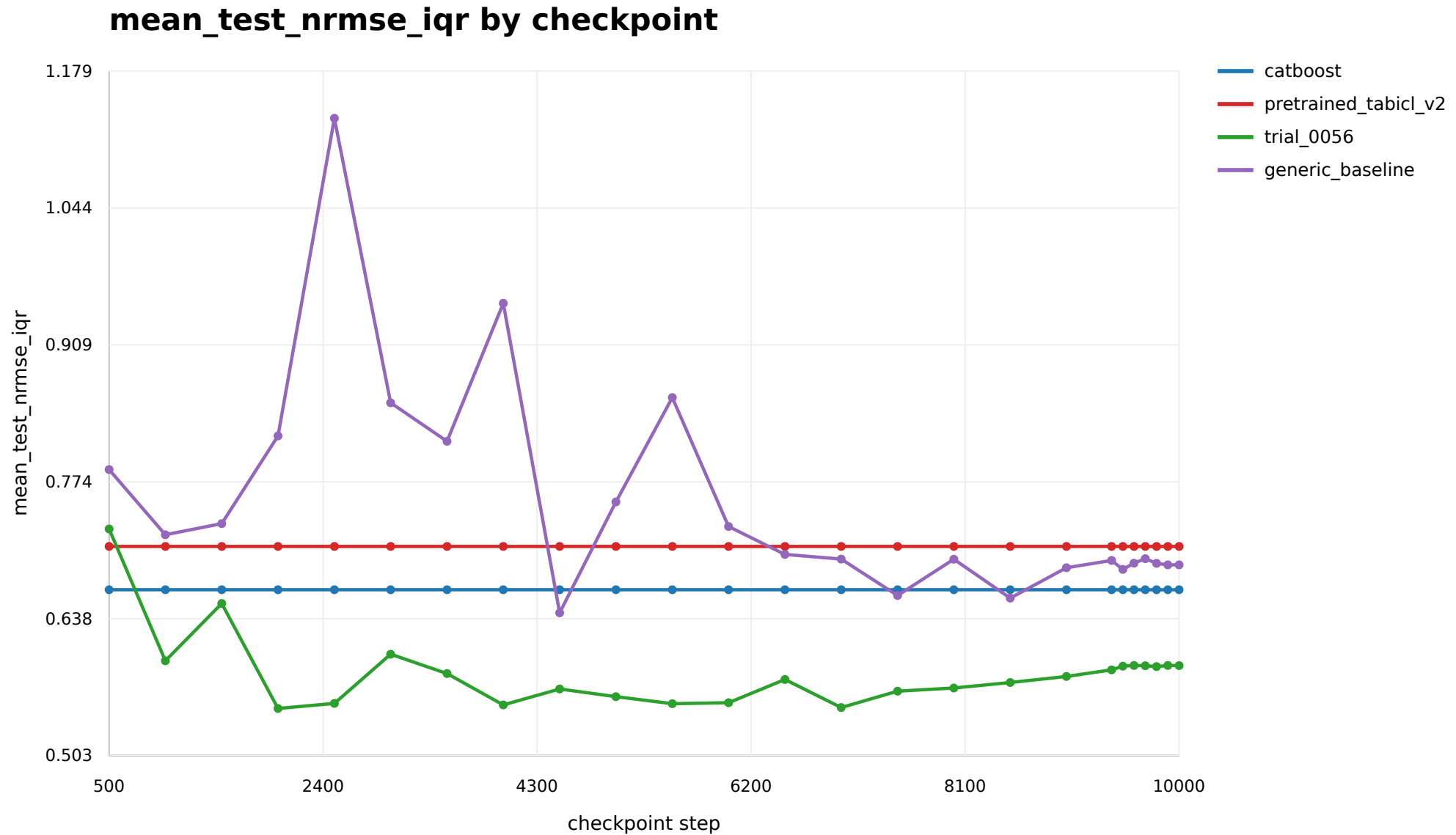
Appendix · 10. Mean Absolute Error (MAE)



Appendix · 11. Normalized MAE (IQR)



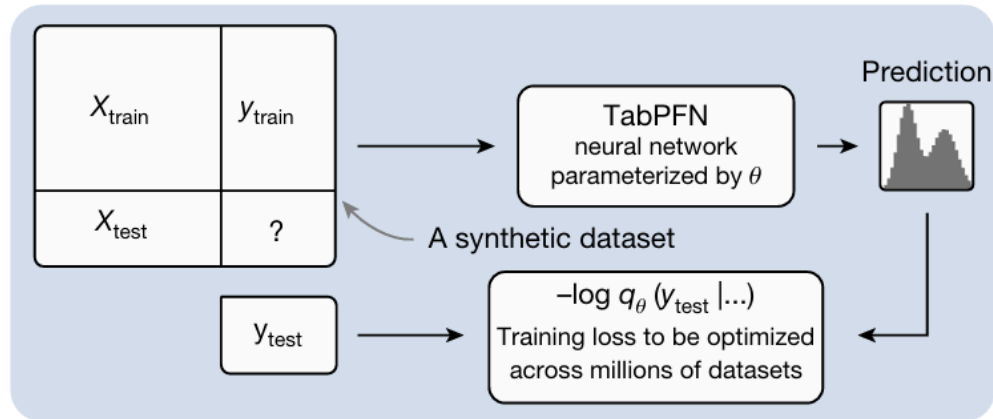
Appendix · 12. Normalized RMSE (IQR)



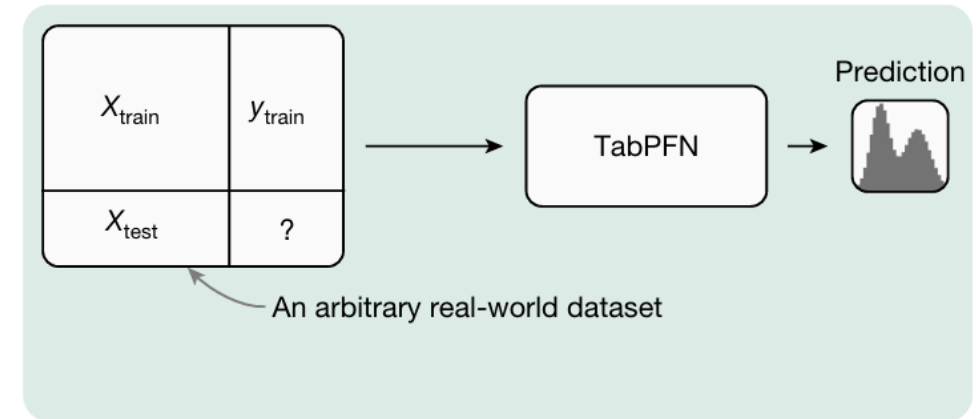
Appendix · 13. TabPFN Training and Architecture

a

TabPFN is trained on synthetic data to take entire datasets as inputs and predict in a forward pass



TabPFN can now be applied to arbitrary unseen real-world datasets



b

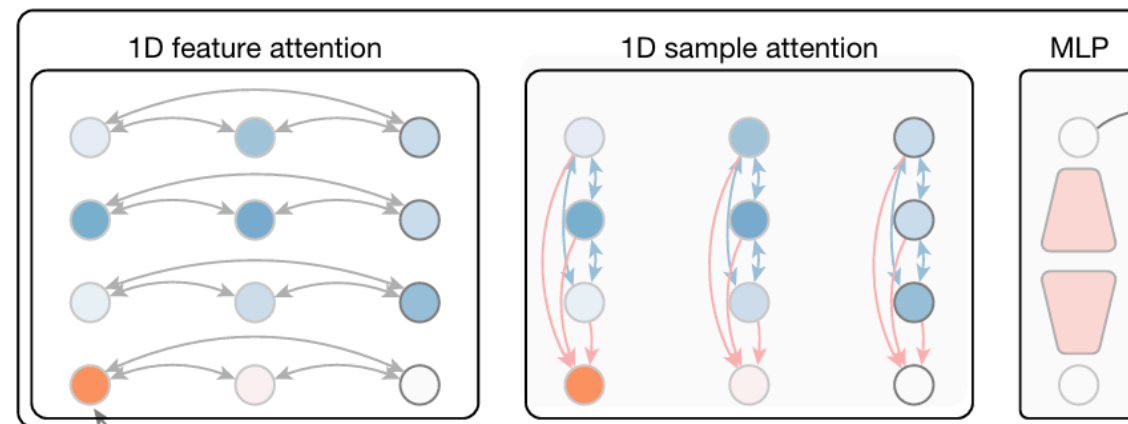
Input dataset

Training rows
Test

x_1	x_2	y
1.2	6.1	3.0
8.9	9.1	3.1
1.0	2.9	6.7
33.3	2.2	?

We predict this entry

2D TabPFN layer (12x)



Predictions: $\hat{y}_{\text{test}}$

The vector is transformed to a piece-wise constant (Riemann) distribution with an MLP

